

- 5 Fig. 5.1 shows a circuit containing a battery, two fixed resistors X and Y, and a light-dependent resistor (LDR) Z.

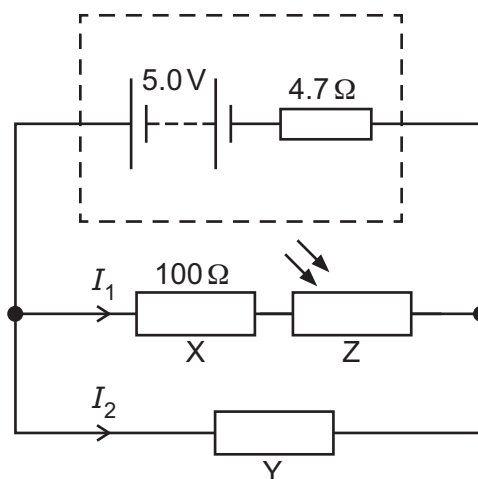


Fig. 5.1

The battery has electromotive force (e.m.f.) 5.0 V and internal resistance 4.7 Ω. The current in X is I_1 and the current in Y is I_2 .

The resistance of X is 100 Ω. The resistance of Z varies with the intensity of light incident on it as shown in Fig. 5.2.

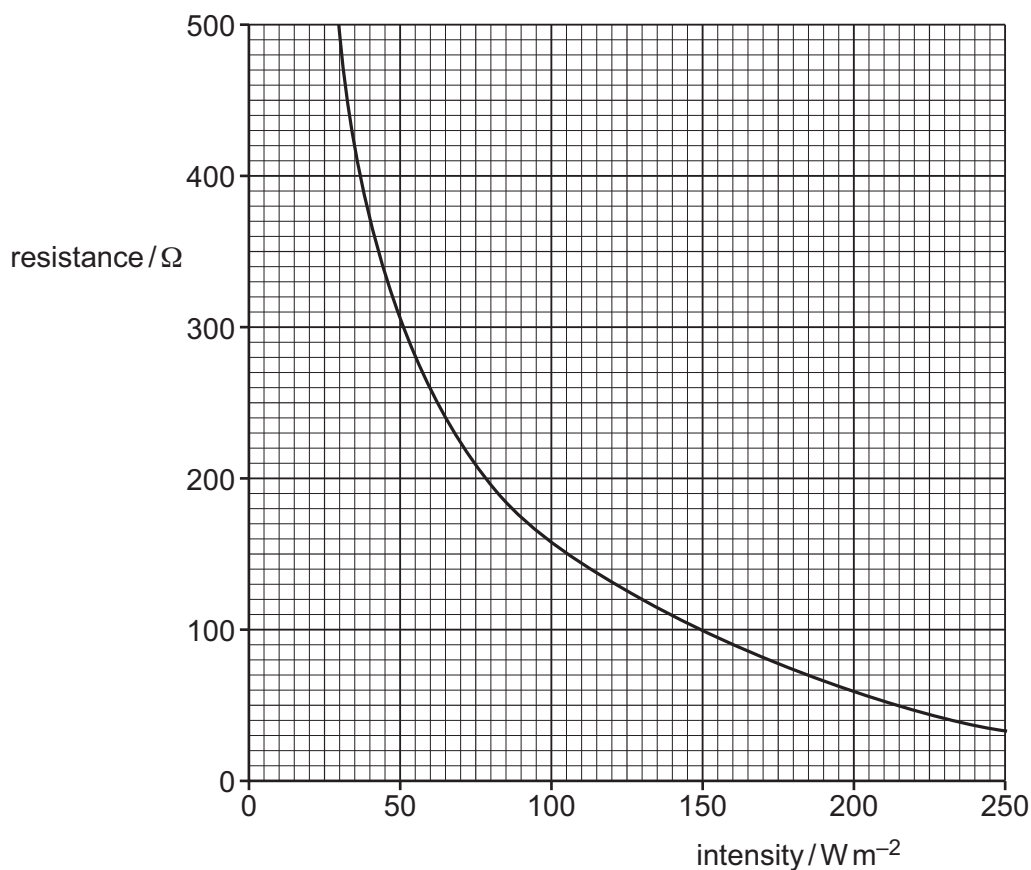


Fig. 5.2



(a) State Kirchhoff's first law.

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 [1]

(b) The intensity of light incident on Z is 130 W m^{-2} . The current in the battery is 38 mA.

(i) Show that the terminal potential difference of the battery is 4.8 V.

[2]

(ii) Calculate the current I_2 in Y.

$I_2 = \dots\dots\dots \text{ A}$ [3]

(iii) Calculate the power dissipated in Y.

power = $\dots\dots\dots \text{ W}$ [2]

(iv) The intensity of the light incident on Z decreases.

State and explain the effect on the terminal potential difference of the battery.

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 [3]

[Total: 11]

